System Design Basics

Lesson 3: Data Storage Choices

Data Storage Choices - Study Notes

Key Concepts

- Relational vs. document/graph stores
- CAP theorem (Consistency, Availability, Partition tolerance)
- ACID vs. BASE properties

Summary

Choosing the right database depends on data model, consistency needs, and scalability. SQL databases excel at complex queries and strong consistency, while NoSQL databases offer flexible schemas and horizontal scaling. The CAP theorem helps decide trade offs: a system can provide at most two of the three guarantees. Understanding ACID (Atomicity, Consistency, Isolation, Durability) versus BASE (Basically Available, Soft state, Eventual consistency) guides design.

Important Points

- Use SQL for relational data with strict consistency.
- Use NoSQL for high write, schema flexible workloads.
- Partition tolerance is essential for distributed systems.

Example

Store user profiles in PostgreSQL with foreign keys for relationships, and use Redis for session caching.

Quick Review

- What does ACID guarantee?
- Which CAP guarantee is usually sacrificed in distributed systems?